

SHAIK ABDUL SAMAD

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EDUCATION

INDIAN INSTITUTE OF TECHNOLOGY DHARWAD (IIT DH) | MTech., Electrical Engineering – Communications, Signal Processing and Machine Learning | 2023-2025 | CGPA: **8.96/10**

IIIT RGUKT RK Valley | B.Tech., Electronics and Communication Engineering | 2019-2023 | CGPA: **8.75/10**

IIIT RGUKT RK Valley | Pre-University Course (MPC) | 2017-2019 | CGPA: **8.87/10**

SKILLS

Programming Languages | C, Python

Technologies | Machine Learning, Deep Learning, Audio Signal Processing, Speech Processing, DSP, Computer Vision, Embedded Systems, Generative AI

Embedded Systems | Tiva C, ARM Cortex-M, Arduino, Code Composer Studio, Arduino IDE

Model Architectures | CNNs, RNNs, Transformers, U-Nets, Diffusion Models, Statistical Models, LLMs (BERT, GPT etc)

Optimization & Deployment | Model Optimization, Compression, Knowledge Distillation, ONNX Runtime

ML/AI Frameworks | TensorFlow, Keras, PyTorch, ONNX, Scikit-learn, Numpy

Version Control & Miscellaneous | Git & GitHub, Shell Scripting, Docker, Multi-threaded programming (Socket Programming)

EXPERIENCE

- **Infobell IT Solutions Pvt Ltd** | Trainee Engineer Intern | Jun 2025 – Till date
 - **Skills:** Agentic AI, RAG, MongoDB, ML Inference Optimization
 - Implemented RAG based system to retrieve benchmark datasets from AMD.
 - Optimizing ML inference pipelines with focus on **latency reduction**.
- **Siemens** | Technical Intern | Nov 2024 - May 2025
 - **Skills:** AI/ML, Federated Learning, Multi-threaded programming, Computer Vision, ONNX Inference
 - Explored **Federated Learning** for domain adaptation: Computer Vision & Time Series Foundation Models.
 - Built **multi-threaded socket communication** between FL clients & servers for parallel execution.
 - Experimented different methods to improve model performance with FL approach by 10-20%.
 - Distributed and Parallel computing to reduce model training time.

PROJECTS

Dysarthria Speech Detection

- Extracted **MFCCs** and **wav2vec2 embeddings** as speech features for classification.
- Implemented **RNNs, LSTM, Autoencoder, and TDNN models** for binary classification (Dysarthria vs. Non-Dysarthria).
- Achieved 96.25% accuracy, demonstrating robustness of deep learning models for speech disorder detection.

Custom C Library for Controlling RGB LEDs on NEO PIXEL JEWEL Module with TM4C123GH6PM Microcontroller

- Utilized a **bit-banging approach** to control RGB LEDs by generating time delays in the order of a few nanoseconds, ensuring **accurate timing and synchronization** for LED operations.
- Implemented advanced LED control functions enhancing the functionality of the NEO PIXEL JEWEL module.
- Demonstrated **real-time signal generation, synchronization**.

Implementation of DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter

- Teamed up with 2 students to replicate the research paper titled "Implementation of **DistilBERT**, a distilled version of BERT: smaller, faster, cheaper, and lighter."
- Implemented the **BERT model** to train the '**Recognizing Textual Entailment (RTE)**' dataset, achieving a score of **64.98**, compared to the original paper's score of **69.3**.
- Implemented both the **DistilBERT** and **CustomInLawBERT** models to train the "**Rhetorical Role Prediction Dataset**" as an additional task beyond what was presented in the original paper.

Master's Thesis: Deep Learning Enabled Segmentation, Classification and Probability Risk Assessment of Cervical Cancer Through Feature Extraction

- Implemented Various Deep Learning Models (CNNs, Vision Transformers) for segmentation task and classification task using **SiPaKMeD** dataset.
- Designed a **novel deep learning architecture (MRF-DCN)** that achieves classification **accuracy around 96%** with fewer learnable parameters compared to existing state-of-the-art models.
- Designed a **Multi-task UNet with a squeezed bottomneck layer** by modifying the **U-Net** model for multi-task prediction.
- Developed an integrated **pipeline for segmentation, classification, and probabilistic risk assessment** to improve early detection and prediction of cervical cancer progression.

Bachelor's Thesis: AI-Powered Embedded System for Fatigue Detection and Relief

- Designed a new embedded system to get relief from tiredness integrating **computer vision** to capture human facial status & multiple components including controlling Fan/AC, playing relaxing music, and deploying a small robot to provide water.
- Gathered **facial expression data** from university students of total 400 to train the YOLO model for fatigue detection.
- Tools used: **Arduino IDE, Python, Bread Board, DC 12V Motors, L298 2A Dual Motor Driver, Relays 2x1, IR Sensor, Image Processing, Deep Learning**

PUBLICATIONS

- Abdul Samad Shaik, Shashaank Mattur Aswatha, Rahul Jashvantbhai Pandya, "Deep Learning Enabled Segmentation, Classification and Risk Assessment of Cervical Cancer," arXiv preprint arXiv:2505.15505, 2025

COURSEWORK

- Speech Processing & its Laboratory
- Pattern Recognition and Machine Learning & its Laboratory
- Neural Networks and Deep Learning & its Laboratory
- Natural Language Processing
- Embedded Systems Design & its Laboratory

ADDITIONAL RESPONSIBILITIES

- **Teaching Assistant** | Embedded Systems Design Lab, IIT Dharwad | *Aug 2024 – Nov 2024*
 - Mentored 50+ students, guiding them in coding for the **Tiva TM4C123GH6PM** microcontroller.
 - Guided students in low-level embedded C coding and real-time control, relevant to microcontroller systems.
- **Robotics Club Coordinator** | Coordinator of Robotics Club of IIIT, RGUKT, RK Valley | *Jan 2022 – Apr 2023*
 - **Led and mentored junior Students** on multiple projects, guiding them through the implementation stages.
 - Conducted comprehensive **teaching sessions on Arduino, NodeMCU, and Machine Learning**
- **Workshop Organizer** | Astronics 4.0 - IIIT, RGUKT, RK Valley | *April 2023*
 - **Taught programming** concepts, enhancing participants' technical skills during the workshop.
 - Assisted students **with hands-on experiments**, fostering a practical understanding of the workshop material.
 - Organized and **managed the entire Astronics-4.0** workshop, ensuring smooth execution of all activities.

CERTIFICATES

- **Coursera** | Neural Networks and Deep Learning by deeplearning.ai | *July 2022*
- **Udemy** | Machine Learning A-Z™: Hands-On Python in Data | *May 2020*
- **Udemy** | Deep Learning A-Z™: Hands-On Artificial Neural Networks | *April 2020*
- **CISCO Network Academy** | Cisco NetAcad – ISTE National Level CODATHON-2020 | *April 2020*

KEY ACHIEVEMENTS

- Secured **1st Prize** in Technical Fest and **awarded Rs. 20,000** conducted by IIIT RGUKT RK Valley | *2023*
- Secured **92.23** percentile in **Electronics and Communication GATE 2023** among 45,833 candidates | *2023*
- Secured **94.93** percentile in **Electronics and Communication GATE 2022** among 54,292 candidates | *2022*
- Runner up in CISCO NetAcad – ISTE **National Level CODATHON-2020** | *2020*
- Secured **98.75** percentile in **Joint Entrance Examination (MAIN) 2019** among 11,47,125 candidates | *2019*

AREA OF INTERESTS

- Audio AI
- Voice AI
- Embedded ML